

DSA Day 33: Breadth First Search (BFS)

Book: Data Structures Through C in Depth - Ch7 §7.12.1 · pp 353-363 | Code in Python

Overview

BFS explores level by level using a queue; gives shortest paths (fewest edges) in an unweighted graph.

Key concepts to master

- visited set
- queue frontier
- level order / shortest hops

Python implementation

```
from collections import deque
def bfs(g,s):
    seen={s}; q=deque([s]); order=[]
    while q:
        u=q.popleft(); order.append(u)
        for v in g[u]:
            if v not in seen: seen.add(v); q.append(v)
    return order
```

Complexity

$O(V+E)$.

Common pitfalls

- Marking visited on dequeue instead of enqueue (duplicates)
- Forgetting visited entirely

Practice

- BFS from node 0
- Why BFS gives shortest unweighted paths

